Python Assignment
Topic-4: Conditional Statements

Q1. Introduction to conditional statements: if, else, elif.

Ans.

Conditional statements are used to make decisions in a program. They allow the program to run different code based on conditions. The 'if' statement checks a condition first. If the condition is true, the code inside it runs. The 'else' statement runs when the condition is false. The 'elif' statement is used when we have multiple conditions to check. These statements help in controlling the flow of the program. They are very useful in real life programs where decisions are required.

Example:

```
x = 12
if x > 10:
   print("Greater than 10")
else:
   print("Less or equal")
```

Q2. Nested if-else conditions.

Ans.

Nested if-else means using one condition inside another condition. It is useful when we need to check multiple levels of conditions. In this, the inner condition runs only if the outer condition is true. It helps in solving complex problems step by step. Proper indentation is very important in nested conditions. It makes the code clear and easy to understand. Nested if should be used carefully to avoid confusion. It improves logical thinking in programming.

Example:

```
x = 15
if x > 10:
   if x < 20:
      print("Number is between 10 and 20 ")
```